Homework 32: Least Squares Problem I

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BCS A, E, F, G

Problem 1. ***Finding Least Square Solution for a Consistent System.*** Find the least squares solution, the least squares error vector, and the least squares error of the consistent system $A\boldsymbol{x} = \boldsymbol{b}$ for

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Problem 2. ***The Role of QR-Factorization in Least Squares Problems.*** Using $A = QR$, find the unique least squares solution.

Problem 3. ***The Moore-Penrose Pseudoinverse.***

a. Find the Moore-Penrose Pseudoinverse A^{+} of A.

b. Verify that $A^{+}A = I$.

c. Is A^{+} the right inverse of A?

Problem 4. ***Invertible Matrices and The Moore-Penrose Pseudoinverse.*** Find the inverse of matrix A. What is relationship between A^{-1} and A^{+}. Justify your answer.

Problem 5. ***Standard Matrix for Orthogonal Projection.*** Find the standard matrix for the orthogonal projection on the column space of matrix A.

Problem 6. ***In a Nutshell.***

a. Use $A = QR$ to show that the projection matrix $A(A^{T}A)^{-1}A^{T}$ can be written as QQ^{T}.

b. For the given matrices

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

- **i.** Find the projection of $\boldsymbol{b}$ on the column space of A.
- **ii.** Find the QR decomposition for A.

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Good Luck